

GABRIEL ADRIAN MANAS

Data Analyst / Data Scientist

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SUMMARY

Medical physicist and quantitative researcher transitioning into **data science / analytics**, backed by an MS in Applied Physics and a strong foundation in **Python, statistics, and numerical modeling**. Experienced turning measurement, imaging, simulation, and business data into rigorous, decision-ready analysis — and communicating it through dashboards in **Power BI and Tableau**. Also a working full-stack developer, comfortable writing SQL and building data pipelines end to end.

TECHNICAL SKILLS

Programming & Querying	Python, SQL, MATLAB, R (familiar)
Data & Analysis	pandas, NumPy, SciPy, scikit-learn, statsmodels, statistical analysis, numerical methods, Monte Carlo (PHITS)
Visualization & BI	Power BI, Tableau, Matplotlib, Seaborn, Excel, Looker Studio
Databases & Tools	MySQL, PostgreSQL, Jupyter, Git/GitHub, Docker, LaTeX
Engineering (secondary)	Laravel, PHP, React, REST APIs
Domain	Medical imaging QA/QC, dosimetry, radiation safety

EXPERIENCE

Junior Medical Physicist

Apr 2026 – Present

Medical Physics & Health Physics Services, Inc. (MPHPS)

- Capture and analyze **quantitative performance data** across imaging modalities (CT, radiography/fluoroscopy, mammography, PET/CT, SPECT/CT) during QA testing.
- Compare image-quality and radiation-output metrics against regulatory tolerances to verify performance and **flag deviations**, applying statistical QC across repeated measurements.

Data Analyst / Full-Stack Developer

Jun 2025 – Present

Remote Freelance Work

- Built **revenue, MRR, and churn reporting** for a subscription product; wrote SQL across billing and subscription tables to surface weekly finance and ops metrics.
- Designed **dashboards and KPI views (Power BI)** for pricing, add-on adoption, and billing configuration, replacing manual spreadsheet pulls with repeatable queries.
- Owned data validation and reconciliation for invoices and client sync; built Laravel/React features supporting the reporting workflows.

Science Teacher (Elementary – Senior High)

Jun 2025 – Apr 2026

Hephzibah School Inc.

- Taught science and quantitative reasoning, translating statistical and analytical concepts into clear, accessible explanations — sharpening data-communication skills.

Data & Backend Developer

Dec 2024 – May 2025

Beth Aven — E-Commerce

- Wrote **SQL for sales, inventory, and order analytics**, giving the team visibility into product performance and stock movement.
- Built ETL-style flows from operational tables into clean, reporting-friendly views.

Medical Physics Intern

May – Jun 2024

UP – Philippine General Hospital (PGH)

- Collected, encoded, and validated **patient dose data** and performed activity/dose calculations across diagnostic radiology, nuclear medicine, and radiation oncology.
- Conducted radiation surveys and compiled QA/compliance documentation across X-ray, CT, MRI, SPECT/CT, PET/CT, brachytherapy, and LINAC systems.

Junior Developer <i>HPL Game Design</i>	2025
Built responsive, data-driven UI components and instrumented user-interaction tracking feeding engagement views.	
Full-Stack Developer Intern <i>NexVision</i>	2023
Built SQL-backed reporting features and database-driven views; assisted with data cleanup and API integration.	

PROJECTS & RESEARCH

Undergraduate Thesis — Numerical Evaluation of Bragg–Gray Cavity Integrals <i>Python, MATLAB</i>	2025
Built numerical models for stopping-power ratios in radiation dosimetry; modeled mass electronic stopping power as a Gaussian and evaluated cavity integrals by numerical integration.	
PHITS Monte Carlo Simulation <i>Python, PHITS</i>	2025
Ran radiation-transport simulations (1M+ histories) and analyzed output to determine Bragg–Gray stopping-power ratios and validate the analytical model.	
Revenue & Subscription BI Dashboard <i>Power BI, SQL, DAX</i>	2025
MRR, churn, retention, and ARPU from modeled SQL views; replaced weekly manual spreadsheets with a single live report.	

RESEARCH PRESENTATIONS

- 43rd Samahang Pisika ng Pilipinas Physics Conference (2025)** — “Numerical Evaluation of Bragg–Gray Cavity Integrals by Modeling the Mass Electronic Stopping Power as a Gaussian Function.”
- UST Research Center for Natural and Applied Sciences Symposium (2025)** — “Numerical Evaluation of Bragg–Gray Cavity Integrals.”
- UST Department of Mathematics and Physics Seminar (2025)** — “Using PHITS to Determine the Bragg–Gray Stopping Power Ratio.”

EDUCATION

M.S. Applied Physics (<i>Medical Physics</i>) — University of Santo Tomas	2026 – Present
B.S. Applied Physics (<i>Instrumentation</i>) — University of Santo Tomas	2021 – 2025